# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The sudden drop in pressure at the centre of a tornado can cause houses to: 1

**(a)** stay intact **(b)** get magnetised **(c)** become cold **(d)** burst outwards

**2.** Assertion (A): Atmospheric pressure decreases with height. Reason (R): The amount of air above decreases as we go up. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**3.** The air pressure in a car tyre is measured at 200 kPa. The force on 0.01 square metre of the tyre wall is: 1

**(a)** 2000 N **(b)** 20000 N **(c)** 200 N **(d)** 20 N

**4.** The windward side of a mountain has more rainfall because: 1

**(a)** moist air rises and cools there **(b)** air sinks there
**(c)** it has no clouds **(d)** it is nearer to the Sun

**5.** Assertion (A): The water jet from the lowest hole of a bottle goes farthest. Reason (R): Pressure of liquid increases with depth. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**6.** In a barometer the reading at a hill station is lower than at sea level because: 1

**(a)** the barometer is faulty **(b)** hill air is colder
**(c)** hills attract less **(d)** there is less air above the hill station

**7.** A person climbing Mount Everest carries oxygen because: 1

**(a)** it is warm **(b)** air is dense
**(c)** oxygen is heavy **(d)** air pressure and oxygen are low

**8.** Assertion (A): Thunderstorm clouds develop mostly in warm, moist air. Reason (R): Warm moist air rises quickly and cools to form clouds. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**9.** **Assertion (A):** A cyclone weakens after it reaches the land. 1
**Reason (R):** Land supplies more warm moist air to a cyclone than the ocean does.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** Winds blow from a region of high pressure to a region of low pressure. 1
**Reason (R):** Air moves so as to reduce the difference in pressure.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Why is the eye of a cyclone dangerous when it passes, though it appears calm? 2

**12.** The cabin of an aeroplane has some pressure of 80 kPa while outside at height it is 25 kPa. Find the net outward pressure on the window of 0.1 square metre area and the force. 2

**13.** A tank has water 4 m deep. Find how the pressure at the base compares with that at 1 m depth. 2

**14.** Explain why the water in a dam stays contained in a thick curved wall. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Explain the monsoon winds' reversal in direction between summer and winter in terms of land-sea heating. 3

**16.** Why are the walls of a submarine very strong? 3

**17.** Explain the changes in the weather before, during and after a thunderstorm. 3

OR

A weather report says the pressure in the centre of a cyclone is 95 kPa and the outside pressure is 101 kPa. Find the difference and explain its effect. 3

**18.** A force of 12 N acts on 3 square cm. Find the pressure in N per square cm and in Pa (1 square cm = 0.0001 square m). 3

OR

Why must we avoid using electrical appliances and a landline phone during lightning? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A water tank at the top of a building has a tap at the bottom floor and another at the third floor. Water comes out with a strong flow from the bottom tap and only a weak flow from the upper tap. 5
(i) Which tap has higher water pressure?
(ii) Name the factor on which liquid pressure depends.
(iii) Does water press only downwards on the tank?
(iv) Explain the strong flow from the bottom tap.

**20.** Explain why wind speeds increase and why the sea breeze moves inland during the day and compare with the land breeze at night, using pressure and temperature differences. 5

OR

Plan a school-level campaign to prepare students and families for storm season, listing five activities. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**